Nebula9 Enterprise Knowledge Intelligence Reference Platform

Detailed PRD • Manufacturing / Industrial first implementation • Retail & CPG portability validation

Document field	Definition
Version / status	Expanded working baseline / for stakeholder review
Audience	Nebula9 leadership, CXO prospects, solution architects, engineering and project intern
Product posture	Demonstration/reference platform with production-grade architecture and a clear path to production

Core proposition

Do not build another generic chatbot. Demonstrate how an enterprise can ask one business question across disconnected knowledge sources, have specialized AI agents retrieve and reason over the right information, validate evidence, and return a traceable answer.

1. Executive Summary

Nebula9 has previously built multi-agent retrieval-augmented generation (RAG) capability for enterprise environments. The underlying client implementations are covered by non-disclosure agreements. This initiative therefore creates a sanitized reference implementation using synthetic and/or approved public data; it must not expose, copy, or recreate any client-specific implementation, data, screenshots, or proprietary configuration.

Externally, the platform is positioned as Enterprise Knowledge Intelligence. “Multi-Agent RAG” describes an implementation mechanism, not the primary sales proposition. The business value is to make fragmented enterprise knowledge usable in context: a business leader asks a consequential question, the platform gathers evidence from relevant sources, validates it, and returns a concise answer that can be inspected.

Manufacturing / Industrial is the first complete scenario because it naturally combines operational documents, structured production and quality data, and controlled external intelligence. Retail & CPG is the second configuration used to validate reusability. Additional industry packs may follow after the shared foundation is demonstrated.

The intended demonstration

- Ask a realistic question that cannot be answered reliably from a single source.
- Observe the system select and invoke specialized agents appropriate to the question.
- Review a consolidated answer with claim-level evidence, source traceability and metrics.
- See conflicts and insufficient evidence handled explicitly rather than hidden.
- Switch industry context without rebuilding the core platform.

Success statement	A non-technical executive should understand the problem, the connected knowledge types, the answer, and the evidence behind it within 60–90 seconds.

2. Product Vision, Positioning & Principles

2.1 Product vision

Turn fragmented enterprise knowledge into trusted, contextual answers that support faster research and better-informed decisions. The platform is a reusable demonstration of an enterprise knowledge layer, not a replacement for source-of-record systems and not an autonomous decision-maker.

2.2 Product hierarchy

Layer	Definition / Responsibility
1. Business proposition	Enterprise Knowledge Intelligence: contextual answers from fragmented knowledge.
2. Core platform	Multi-agent orchestration, retrieval, structured data access, controlled external intelligence, evidence validation and synthesis.
3. Enterprise controls	Authentication, RBAC, workspace/source permissions, auditability, governance, observability and security.
4. Industry configuration	Manufacturing, Retail & CPG, Technology & Services, Insurance, Healthcare & Life Sciences, Financial Services.
5. Business use case	Operations, supply chain, quality, research, risk, compliance, customer intelligence and decision support.

2.3 Product principles

• Evidence before eloquence: answer quality is subordinate to source support.

• Modular by design: industry-specific data, terminology, scenarios and evaluation sets are configuration.

• Least privilege: source access and tool permissions are enforced server-side.

• Visible uncertainty: conflicting or missing evidence is a first-class response state.

• Executive-first UX: technical detail is available on demand, not forced on every user.

• Production-minded, demo-scoped: implement real boundaries without overbuilding unnecessary infrastructure.

2.4 Explicit non-goals

The first release is not a general-purpose chatbot, not a client-specific reproduction, not a blanket claim of production readiness for regulated environments, and not a collection of unreviewed AI-generated code.

3. Business Problem & Opportunity

Enterprise knowledge is distributed across documents, operational databases, team repositories and external information. Teams manually reconcile these sources, repeat research, wait for answers, and struggle to verify the provenance of claims. Search alone may retrieve relevant material but often leaves the user to perform cross-source reasoning.

3.1 The gap to demonstrate

Current state	Desired state
Knowledge siloed by system and format	One query can retrieve from approved heterogeneous sources
Manual cross-checking and synthesis	Specialized agents gather source-specific findings
Answer separated from its evidence	Claims link to inspectable source passages or data context
Disagreement hidden or casually resolved	Conflicts are surfaced with dates, versions and uncertainty
Industry-specific prototype requires rewrites	Industry packs change through configuration and data replacement

3.2 First manufacturing scenario

A plant leader asks: “What are the major factors contributing to production downtime at Plant A, and what actions have previously reduced similar downtime?” A credible response may require maintenance manuals, downtime records, quality reports, supplier documents and approved external context. The platform must synthesize the evidence without implying that correlation proves causation.

3.3 Opportunity boundaries

The reference platform should make the capability tangible to prospects and provide a credible technical foundation for discovery and future production engagements. Business impact such as reduced research time or improved operational outcomes is a hypothesis to validate in customer-specific pilots, not a pre-claimed outcome.

4. Goals, Non-Goals & Success Criteria

4.1 Product goals

• Deliver an end-to-end web application for a multi-source business question.

• Demonstrate document, structured-data and controlled external retrieval.

• Show agents, evidence, traceability, conflicts and insufficient-evidence behavior.

• Implement real authentication, authorization, audit and observability controls where feasible.

• Separate industry configuration from shared application logic.

• Validate the same core platform with Manufacturing and Retail & CPG packs.

• Package a polished CXO demonstration and technical overview.

4.2 Success criteria

Dimension	Acceptance signal
End-to-end function	A representative cross-source question completes from query to answer.
Grounding	Important material claims have supporting evidence or are explicitly qualified.
Traceability	A user can inspect source identity and location/context for a cited claim.
Multi-agent behavior	Agents are selected for relevant source types and their work is observable.
Security	Authentication, authorization and tool/source boundaries are enforced server-side and tested.
Modularity	Both industry packs run on the same core without a platform rewrite.
Quality	Unit, integration, security, evaluation and smoke tests exist and critical scenarios pass.
Performance	Latency is measured on this implementation and reported honestly.
UX	A non-technical executive understands value and evidence flow quickly.
Sales readiness	A repeatable 3–5 minute story demonstrates the business problem and product behavior.

Measurement guardrail Historical metrics from an earlier case study must not be reused as performance claims for this reference platform unless independently measured.

5. Users, Personas & Stakeholders

5.1 Primary personas

Persona	Primary need	Product response
CXO / business sponsor	Understand value and decision relevance quickly	Executive summary, key findings, business-language agent activity and clear caveats
Operations / domain leader	Investigate a cross-functional operational question	Metrics, evidence from multiple source types, source comparison and follow-up questions
Analyst / knowledge worker	Verify facts and continue research	Citations, passages, dataset context, retrieval metadata and conversation context
Platform administrator	Control access and manage source packs	User roles, source metadata, workspace configuration and audit visibility
Solution architect	Assess integration and production path	Architecture, service boundaries, configuration model, controls and deployment-gap notes
Engineering / evaluator	Diagnose behavior and improve quality	Execution traces, tests, evaluation cases, errors and latency telemetry

5.2 Jobs to be done

• When I ask a cross-source business question, help me see the most relevant supported findings without manually reconciling each system.

• When I need to trust an answer, let me inspect the evidence behind individual claims.

• When sources disagree, show the disagreement and explain only what the evidence supports.

• When I demonstrate the platform to another industry, let me switch data and terminology without rewriting core logic.

5.3 Stakeholder responsibilities

Product ownership defines the demo story and acceptance criteria; engineering owns architecture, implementation and tests; security review validates access and AI-specific threats; domain reviewers validate synthetic scenarios and terminology; sales/demo owners maintain the narrative and prospect-safe collateral.

6. Scope & Release Boundaries

6.1 Release 1 — required

• Executive-first query and answer interface with suggested scenario questions.

• Industry/scenario selector and configuration loader.

• Synthetic/public document registration and retrieval.

• Structured data via CSV and/or PostgreSQL, using approved schemas and read-only access.

• Controlled external-source adapters with explicit allowlists.

• Router/planner, document agent, structured-data agent, external agent, evidence validator and synthesis agent.

• Evidence panel, citations, conflict states and insufficient-evidence behavior.

• Authentication, demonstrable user/admin distinction, server-side authorization and source/workspace boundaries where applicable.

• Audit events, execution traces, latency/error telemetry and an evaluation set.

• Manufacturing pack and Retail & CPG validation pack.

• Setup/runbook, architecture diagram, technical overview and 3–5 minute demo assets.

6.2 Out of scope unless separately approved

• Unrestricted web browsing or arbitrary external API access.

• Production-grade high availability, multi-region disaster recovery or enterprise-scale throughput guarantees.

• Autonomous write operations in operational systems.

• Sector-specific certification or regulatory compliance claims.

• Use of real client data or NDA-protected implementation artifacts.

• Automatic resolution of conflicting records without evidence or an approved policy.

6.3 Production-readiness posture

The implementation should use clear interfaces, secure defaults, configuration separation, tests and observable behavior. A production engagement would still require customer-specific identity integration, deployment architecture, data governance, privacy review, threat modeling, capacity planning, operational support and regulatory assessment.

7. Primary User Journeys

7.1 Journey A — ask and verify

1. User selects Manufacturing and opens a suggested scenario.

2. User submits a business question in natural language.

3. System presents a concise interpretation and begins a traceable execution.

4. Relevant agents retrieve permitted evidence from configured sources.

5. Evidence validation checks claim support, source freshness and conflicts.

6. Synthesis produces an executive-friendly answer with citations and caveats.

7. User opens a citation to inspect the source passage or data context.

8. User asks a follow-up; relevant context is retained without widening access.

7.2 Journey B — inspect a conflict

1. A question returns two material claims that disagree.

2. The answer marks the affected claim as conflicted or unresolved.

3. The evidence panel displays both sources, timestamps/versions and their context.

4. The platform explains a discrepancy only when metadata or source content supports it.

5. The user can continue investigation without the system silently choosing a winner.

7.3 Journey C — switch industry

1. Administrator or demo operator selects Retail & CPG.

2. The configuration loader activates that pack’s labels, source definitions, scenarios and evaluation set.

3. The same shared query, orchestration, evidence and audit services remain in use.

4. A Retail & CPG question demonstrates the same evidence-first pattern with different data.

7.4 Failure journey

If a source is unavailable, access is denied, retrieval is empty, or a tool times out, the system should show a bounded and comprehensible failure state. It must not fabricate successful retrieval, suppress a material source failure, or present unsupported conclusions as verified.

8. Functional Requirements: Query & Answer

Requirements below are intended as testable product behavior. Priority labels: P0 = required for the first complete demonstration; P1 = valuable if time permits; P2 = future enhancement.

ID / priority	Requirement	Acceptance criteria
FR-QA-01 / P0	Business query input	User can enter a natural-language question and submit it without technical query syntax.
FR-QA-02 / P0	Suggested scenarios	Industry pack exposes curated example questions; selecting one populates or submits it predictably.
FR-QA-03 / P0	Intent interpretation	System records a concise interpreted intent and selected source/agent plan for the query.
FR-QA-04 / P0	Structured answer	Answer can show summary, findings, evidence, metrics, traceability, grounding/confidence and conflicts where relevant.
FR-QA-05 / P0	Fact vs inference	Retrieved facts and model-generated synthesis/inference are visually or textually distinguishable.
FR-QA-06 / P0	Follow-up context	Follow-up questions can use relevant prior context while rechecking authorization on each retrieval.
FR-QA-07 / P0	Insufficient evidence	When support is inadequate, system states the limitation and avoids invented specifics.
FR-QA-08 / P1	Answer feedback	User can flag an answer as helpful/unhelpful and optionally provide a reason.
FR-QA-09 / P1	Export/share	Provide a safe way to copy or export answer and citations, subject to access policy.

8.1 Answer presentation contract

The answer should lead with the decision-relevant conclusion, then provide findings and evidence. It should not lead with internal model terminology. Citations should be close to the claims they support. Metrics should include units, time period, population or denominator when available. If those details are missing from the source, the answer should say so.

9. Functional Requirements: Source Management

ID / priority	Requirement	Acceptance criteria
FR-SRC-01 / P0	Register/upload documents	Authorized admin can add approved synthetic/public files with source metadata.
FR-SRC-02 / P0	Document metadata	Record source name/type, version or date, access classification and stable identifier.
FR-SRC-03 / P0	Structured source	Connect CSV and/or PostgreSQL through a constrained adapter.
FR-SRC-04 / P0	Controlled external source	Only configured allowlisted APIs/public sources can be queried.
FR-SRC-05 / P0	Industry pack replacement	Changing source definitions and data pack does not require editing shared core logic.
FR-SRC-06 / P0	Source status	Show configured/available/failed status and useful last-ingested or last-checked metadata where available.
FR-SRC-07 / P1	Re-index lifecycle	Support reprocessing a changed document and avoid treating stale chunks as current.
FR-SRC-08 / P1	Source retirement	Retired or replaced sources stop contributing to new answers; history remains auditable as policy allows.

9.1 Source metadata minimum

• Stable source ID and display name.

• Source type: document, structured dataset or external source.

• Industry/workspace association and access classification.

• Version, effective date or retrieval timestamp as applicable.

• Document page/section or structured table/query context where technically possible.

• Ingestion/indexing state, error details and last successful processing time.

9.2 Ingestion expectations

The ingestion path should validate file type and size, extract text and useful metadata, preserve document/page context when available, chunk content using a documented strategy, and index chunks with stable source references. Unsupported or malformed files should fail visibly and safely.

10. Functional Requirements: Evidence & Conflict Handling

ID / priority	Requirement	Acceptance criteria
FR-EV-01 / P0	Claim traceability	Material answer claims link to one or more supporting evidence items where available.
FR-EV-02 / P0	Document citation	Citation identifies source and page/section or passage location where technically possible.
FR-EV-03 / P0	Structured citation	Citation identifies dataset/table/query or equivalent context and relevant time range where available.
FR-EV-04 / P0	External citation	Citation identifies approved external source and retrieval time where applicable.
FR-EV-05 / P0	Support status	Evidence is classified as supported, partially supported, conflicting or insufficient.
FR-EV-06 / P0	Conflict detection	Materially different values/statements are surfaced side by side.
FR-EV-07 / P0	No invented resolution	System does not silently resolve conflict; any explanation must be supported by evidence such as date/version.
FR-EV-08 / P0	Insufficient evidence	Answer is qualified or declined when available evidence does not support a conclusion.
FR-EV-09 / P1	Evidence filters	User can narrow evidence by source type, date/version or other configured metadata.

10.1 Evidence object

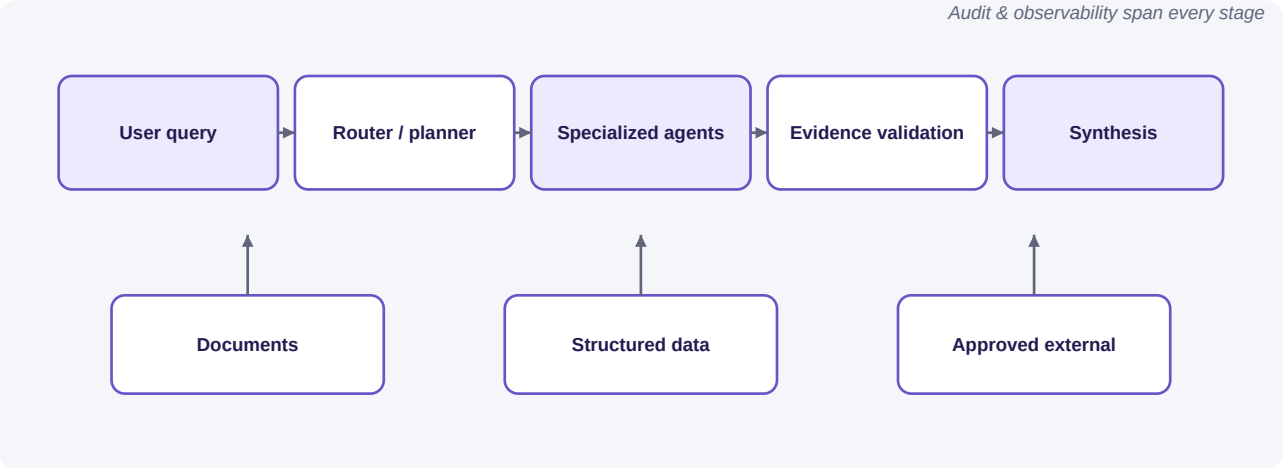
A normalized evidence object should include evidence_id, source_id, source_type, source_title, content or excerpt, location (page/section/table/query), effective date/version if known, retrieval timestamp, access classification, and relevance/support metadata. The final answer should reference stable evidence IDs rather than inventing citation strings.

10.2 Conflict example

A production dashboard may report 14 hours of downtime for a line while a maintenance report lists 11 hours for the same period. The platform should show both values, their source contexts and any available definitions/time windows. It may note a possible scope difference only if the source metadata supports that interpretation.

11. Multi-Agent Architecture

The architecture uses specialized agents for source-specific retrieval and reasoning, with explicit validation before final synthesis. Agents are components with bounded responsibilities and tools; they are not independent authorities that can bypass shared policy enforcement.



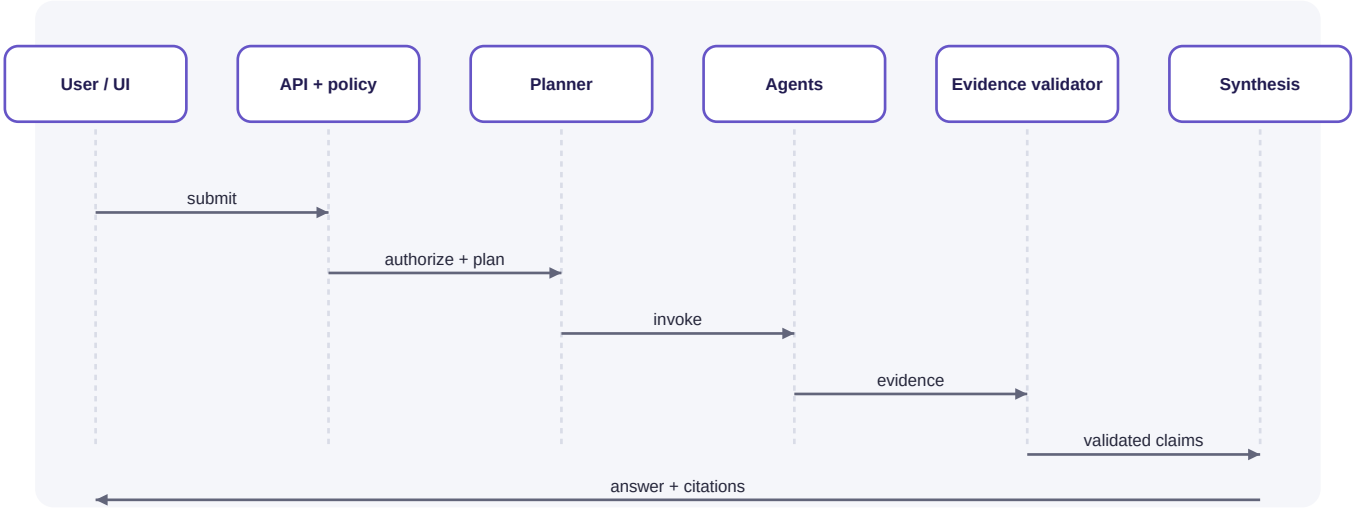
11.1 Core components

Component	Responsibility	Required behavior
Query Router / Planner	Understand intent and plan work	Select likely source types and agents; avoid unnecessary calls.
Document Intelligence Agent	Retrieve/reason over documents	Return passages, source identifiers and page/section context.
Structured Data Agent	Query structured data safely	Use approved schemas and read-only operations; return data and query context.
External Intelligence Agent	Retrieve approved external information	Use allowlists and distinguish external from internal/synthetic evidence.
Evidence / Validation Agent	Assess support and conflicts	Identify supported, partial, conflicting and insufficient claims.
Synthesis Agent	Compose final answer	Combine validated findings while preserving attribution and uncertainty.
Audit / Observability Layer	Record behavior	Capture query, agents, sources, tool/model metadata, latency, errors and outcomes.

11.2 Orchestration principles

- Use typed inputs/outputs between components.
- Propagate a request ID, user/workspace identity and policy context through every stage.
- Treat retrieved content as untrusted data, never as executable instruction.
- Apply timeouts, bounded retries and per-agent error states.
- Do not let synthesis cite evidence that was not returned in the current authorized evidence set.

12. Request Lifecycle & Sequence



12.1 Lifecycle stages

1. Authenticate request and resolve user, role, workspace and active industry pack.

2. Validate question size, session context and applicable policy.

3. Router creates a plan naming expected source types and bounded tools.

4. Agents retrieve permitted evidence in parallel where safe and useful.

5. Normalize evidence and apply source/claim validation.

6. Detect conflict, stale-source conditions and missing evidence.

7. Synthesis creates a structured answer tied to evidence identifiers.

8. Persist audit/telemetry events and return answer plus citations to the UI.

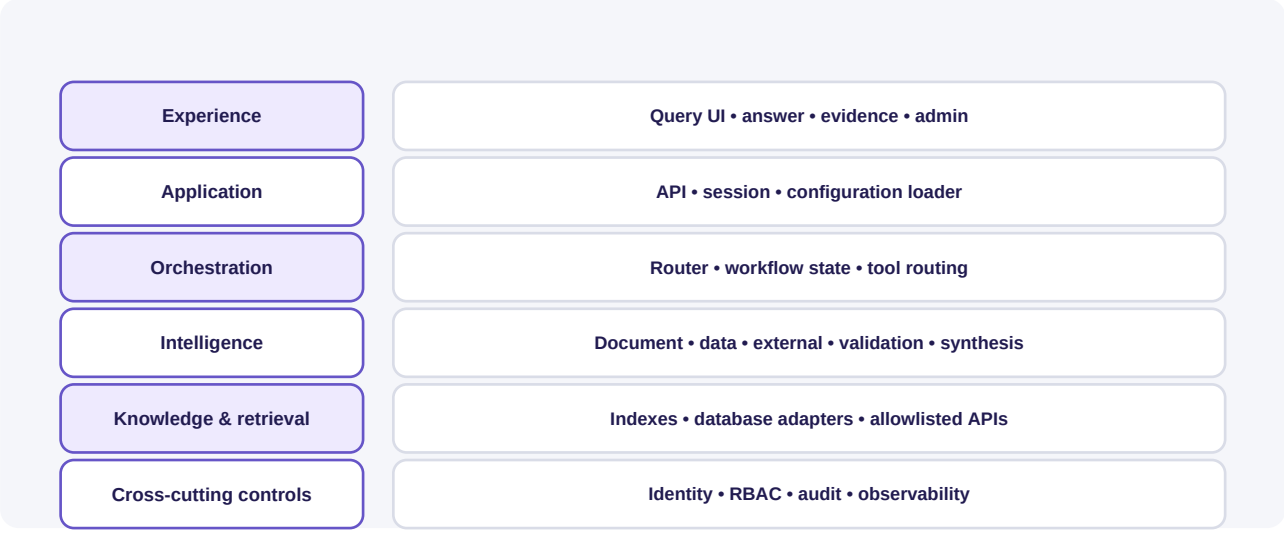
12.2 Orchestration state

A query execution record should track request_id, query_id, user/workspace context, industry pack version, plan, agent statuses, source calls, evidence IDs, validation findings, final answer status, timings, token/usage metadata where available, and error codes. Sensitive payloads should be minimized or redacted according to logging policy.

12.3 Partial failure

The system may return a qualified partial answer if a non-critical source fails and the remaining evidence supports a bounded conclusion. It must disclose the missing source and avoid implying that all configured sources were successfully consulted.

13. Logical System Architecture



13.1 Layer responsibilities

Layer	Primary responsibilities
Experience	CXO query UI, answer view, evidence panel, agent activity, industry selector and admin/source management.
Application	API layer, session/context management, scenario handling, configuration loader and request validation.
Orchestration	Query planner, agent execution, tool routing, workflow state and bounded parallelism.
Intelligence	Document, structured-data and external agents; evidence validator; synthesis agent.
Knowledge / retrieval	Document index/vector retrieval, metadata filters, structured database access and external adapters.
Governance	Authentication, RBAC, source/workspace permissions, audit trail and policy enforcement.
Observability	Execution logs, agent/tool traces, latency, errors and evaluation telemetry.
Configuration	Industry packs, sources, terminology, scenarios, prompts and evaluation datasets.

13.2 Boundary requirements

• UI must not be the enforcement point for authorization.

• Agents call adapters through a policy-aware tool boundary.

• Industry configuration is loaded and validated separately from shared business logic.

• Source adapters expose consistent contracts and normalized evidence.

• Model provider, embedding model and vector-store implementation should be replaceable behind interfaces where practical.

13.3 Deployment note

The reference architecture is logical, not a mandate for microservices. A modular monolith or a small set of services may be appropriate for the demo, provided component boundaries and contracts remain clear.

14. Data Architecture & Information Model

14.1 Core entities

Entity	Key fields / purpose
User	user_id, identity reference, role, status; establishes actor context.
Workspace	workspace_id, name, industry_pack_id, access policy.
IndustryPack	pack_id, version, labels, source definitions, scenarios and evaluation configuration.
Source	source_id, workspace/pack, type, name, version/date, classification, status.
Document / Chunk	document_id, chunk_id, text, metadata, page/section, index version and source linkage.
QueryExecution	request_id, query, actor/workspace, plan, status, timestamps and outcome.
AgentRun	agent_run_id, agent type, input/output references, status and timing.
Evidence	evidence_id, source reference, excerpt/value, location, retrieval timestamp and support status.
Answer / Claim	answer_id, claim_id, text, evidence references, qualification and conflict status.
AuditEvent	actor, action, target, timestamp, outcome and correlation identifiers.

14.2 Data handling principles

• Use stable IDs for sources, evidence and execution traces.

• Retain source version/date and ingestion/index version to make answers reproducible enough for debugging.

• Avoid storing unnecessary sensitive query content in telemetry.

• Define retention and deletion behavior before introducing real customer data.

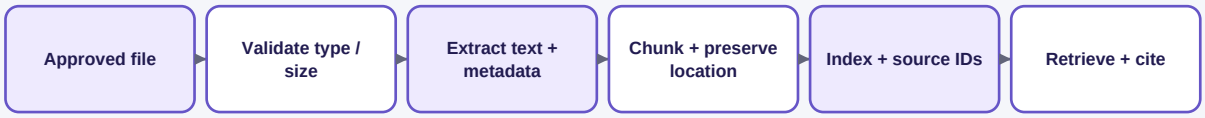
• Treat synthetic demo data as clearly labeled so screenshots and exported collateral cannot be mistaken for production customer data.

14.3 Data quality

Structured data must be checked for required columns, types, null patterns, date ranges and key consistency. Document ingestion should record extraction failures and preserve provenance. The platform should avoid false precision when source values are missing, stale or use incompatible units.

15. Retrieval & Knowledge Processing

15.1 Document pipeline



Every chunk retains provenance to its source and location where available.

15.2 Retrieval requirements

• Apply workspace, source and classification filters before evidence is passed to the model.

• Return source metadata and location with every retrieved passage.

• Use a documented chunking strategy that preserves enough context for interpretation.

• Support hybrid retrieval (keyword and semantic) where feasible; do not assume vector similarity alone proves relevance.

• Deduplicate overlapping chunks and favor source-version consistency.

• Record retrieval parameters and index version for evaluation and debugging.

15.3 Structured data access

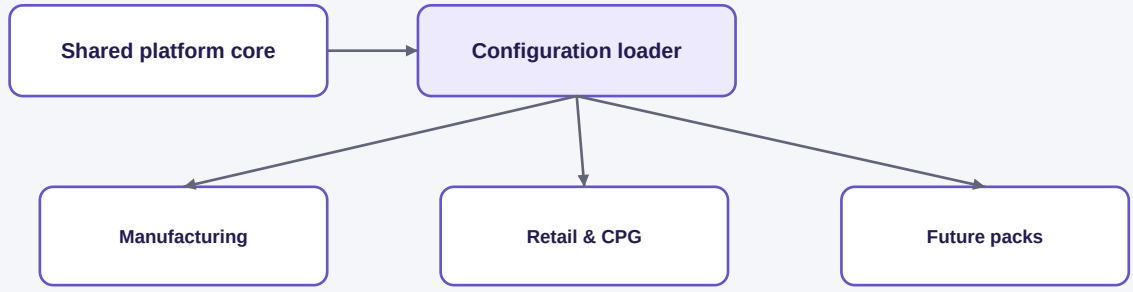
Structured queries must be read-only and constrained to approved tables, columns and operations. Prefer parameterized queries and predefined query templates for common metrics. If natural-language-to-SQL is used, validate the parsed statement and enforce limits, timeout, row caps and schema allowlists before execution.

15.4 External intelligence

External retrieval must use explicit source/API allowlists, bounded requests and a clear distinction between external evidence and internal or synthetic data. Record the source and retrieval time. External content is untrusted and must not change system policy.

16. Industry Configuration Model

Industry packs provide the terminology, data, sources, scenarios and evaluation material that make the shared platform relevant to a domain. They must not fork the core orchestration, access control, evidence validation or audit logic.



Data + configuration change; shared core remains stable.

16.1 Suggested configuration structure

```
/config/industries/manufacturing/\n  industry.yaml\n  terminology.yaml\n  agents.yaml\n  sources.yaml\n  scenarios.yaml\n  prompts.yaml\n  evaluation.yaml
```

16.2 Configuration versus code

Configurable per industry	Core platform code
Industry name and terminology	Agent framework/orchestration
Synthetic/public datasets and documents	Authentication and authorization
Business scenarios and sample questions	Evidence validation logic
Source definitions and metadata	Audit and observability
Industry labels and UI content	Security controls and access enforcement
Evaluation questions and expected evidence	Shared APIs and data contracts

16.3 Pack validation

A pack must pass schema validation, source connectivity checks, a smoke question set and its evaluation suite before it can be enabled. Invalid configuration should fail clearly and must not silently fall back to another industry's sources.

17. Industry Pack 1: Manufacturing / Industrial

17.1 Purpose

Provide the first complete end-to-end scenario around production operations, quality, maintenance, supplier performance and inventory. Use synthetic or approved public material only.

17.2 Illustrative source inventory

Source family	Example contents	Question contribution
SOPs and maintenance manuals	Procedure steps, equipment intervals, revisions	What actions are recommended and what changed?
Production reports	Output, line, shift, downtime hours	Where and when is downtime concentrated?
Quality reports	Defects, inspection results, corrective actions	Does downtime coincide with quality issues?
Supplier documents	Delivery/quality incidents, corrective-action notes	Which suppliers are associated with events?
Inventory/procurement data	Stock levels, lead times, purchase orders	Could part availability be relevant?
Approved external sources	Public supplier/market context or synthetic external records	What external context is relevant and current?

17.3 Core demo questions

• What are the major factors contributing to production downtime at Plant A, and what actions have previously reduced similar downtime?

• Which production lines show rising downtime and declining quality, and what evidence explains the pattern?

• Which suppliers are associated with the highest quality-impact events, and what do supplier documents say about corrective actions?

• What changed between previous and current maintenance procedures, and is there evidence the change affected downtime?

• Where do the production report and maintenance documentation disagree, and which source is more recent?

17.4 Answer guardrails

Do not state causation solely from correlation. State time windows, units and denominators where available. Surface source discrepancies. Distinguish documented recommended actions from actions proven effective in the supplied data.

18. Industry Pack 2: Retail & CPG Validation

18.1 Purpose

Validate that the shared architecture is reusable by replacing Manufacturing data, vocabulary, source definitions and scenarios. This is a portability test, not a requirement to build a second product.

18.2 Illustrative knowledge domains

Source family	Illustrative contents
Sales data	Product, region, period, units, revenue and promotional activity
Inventory data	On-hand stock, stock-outs, replenishment and inventory pressure
Consumer research	Survey summaries, qualitative findings and product feedback
Promotion plans	Campaign timing, spend and expected mechanics
Market intelligence	Approved public/synthetic market context
Product documentation	Product attributes, launches and category information

18.3 Validation questions

- Why did Product X underperform in Region A despite higher promotional activity?
- Which products show declining sales alongside inventory pressure, and what evidence is available?
- What do consumer research documents indicate about reasons for the performance change?
- Are there conflicts between sales data and market research, and how should the answer qualify them?

18.4 Portability acceptance

- Same query UI, orchestration, evidence model and governance controls are reused.
- Only pack data/configuration and explicitly supported adapter settings change.
- No Manufacturing-specific prompt or data assumption leaks into Retail answers.
- Retail evaluation cases verify grounding, conflict handling and insufficient-evidence behavior.

19. Future Industry Strategy

Industry pack	Role	Illustrative domains	Readiness note
Technology & Services	Future / optional validation	Projects, service operations, product docs, delivery metrics	Validate connectors and operational terminology.
Insurance	Future pack	Policies, claims, underwriting, regulatory knowledge	Requires careful policy-version and access controls.
Healthcare & Life Sciences	Future pack	Research, SOPs, clinical/operational knowledge	Requires domain-specific privacy, safety and deployment review.
Financial Services	Future pack	Risk, regulatory, financial and operational knowledge	Requires additional governance and data controls.

19.1 Expansion decision criteria

• Clear prospect use case and an accountable domain reviewer.

• Defined approved data sources and access classification.

• Industry-specific risk and deployment gap assessment.

• Evaluation set with expected evidence and known conflict/abstention cases.

• Evidence that shared core behavior remains stable under new configuration.

19.2 Guardrail

Manufacturing is first because it supports rich cross-source business storytelling without making the initial demo primarily a jurisdiction-specific compliance exercise. Suitability for regulated industries does not imply readiness for them.

20. UX Principles & Information Architecture

20.1 Experience principles

• Executive-first: plain language, concise summary and progressive disclosure.

• Evidence visible: source links and evidence panel are core, not hidden in logs.

• Activity understandable: use phrases such as “Reviewed production data” and “Checked maintenance reports.”

• Uncertainty honest: confidence/grounding indicators must not imply mathematical certainty without calibration.

• Consistent: same answer structure across industries, with domain labels supplied by configuration.

• Accessible: keyboard navigation, visible focus, readable contrast and meaningful status text.

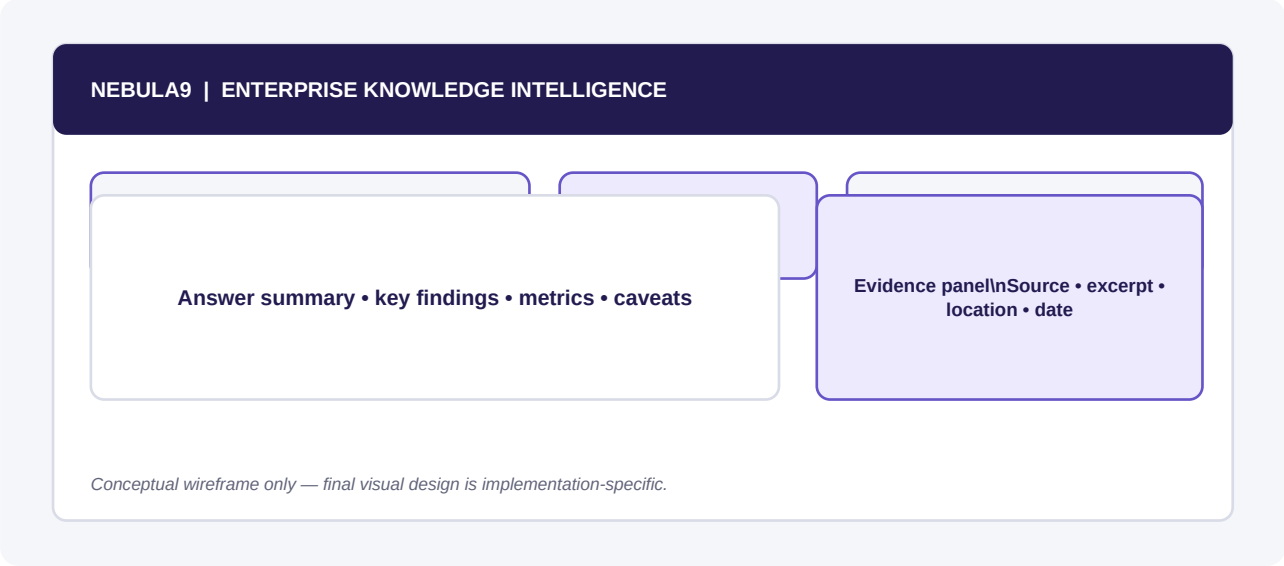
20.2 Recommended information architecture

Area	Primary content
Home / Ask	Industry selector, scenario prompts, query composer and recent activity.
Answer workspace	Executive summary, key findings, metrics, caveats and source-backed claims.
Evidence panel	Source type, title, excerpt/value, page/section/table context, date/version and support state.
Agent activity	Business-language stage status, elapsed time and optional technical details.
Source management	Source registry, metadata, ingestion status and configuration validation.
Admin / governance	Users/roles, workspace/source permissions and audit search, as supported by scope.

20.3 Accessibility and responsive behavior

Core query and evidence workflows must remain usable on common laptop resolutions. Status must not rely on color alone. Evidence drawers and tables should preserve readable content without horizontal overflow where practical. Keyboard users must be able to submit a query, navigate citations and close overlays.

21. UX Wireframe & Answer Anatomy



21.1 Answer anatomy

1. Executive summary: one concise answer to the business question.

2. Key findings: grouped findings with scope and time period.

3. Evidence-backed claims: each material statement tied to evidence IDs.

4. Metrics: value, unit, time range and relevant denominator where available.

5. Conflicts / caveats: discrepancies, missing data, stale sources or assumptions.

6. Source traceability: open source location or data context.

7. Execution details: participating agents, source types, latency and errors on demand.

21.2 Confidence language

Use operational labels such as “Supported by retrieved evidence,” “Partially supported,” “Conflicting sources,” and “Insufficient evidence.” If a numeric confidence score is shown, document how it is computed and evaluated; otherwise avoid presenting a decorative percentage as calibrated probability.

21.3 Empty and error states

Provide clear states for no question entered, no relevant evidence, permission denied, source unavailable, ingestion in progress, unsupported file, timed-out agent, and partial answer. Error copy should explain the next safe action without revealing secrets or internal prompts.

22. Security, Identity & Access Control

22.1 Access requirements

ID	Requirement	Verification
SEC-01	Application authentication	Unauthenticated requests cannot access protected query/source endpoints.
SEC-02	Role-based access control	At least user/admin distinction is demonstrable and tested.
SEC-03	Server-side authorization	Direct API calls cannot bypass UI-hidden permissions.
SEC-04	Workspace/source boundaries	Retrieval filters and tool calls enforce allowed sources.
SEC-05	Secret management	No credentials or API keys in source control, UI or ordinary logs.
SEC-06	Least privilege	Database and external adapters use only required permissions.
SEC-07	Auditability	Sensitive administrative and data-access actions produce audit events.
SEC-08	Demo-data separation	Only synthetic/approved public data is included in demo assets.

22.2 Secure defaults

• Deny access unless policy explicitly allows it.

• Validate identity and authorization at the API/tool boundary on every request.

• Use environment variables or an appropriate secret store; provide a safe example environment file without real secrets.

• Redact tokens, credentials and sensitive configuration from logs.

• Use parameterized database operations and least-privilege database roles.

• Keep development/demo credentials distinct from any future customer environment.

22.3 Production gap

Single sign-on, enterprise identity-provider integration, fine-grained attribute-based policies, key rotation workflows, formal retention controls and customer-specific tenancy may be required for production and should be documented as gaps if not implemented.

23. AI-Specific Threat Model & Guardrails

Threat	Example	Required guardrail / test
Prompt injection in documents	A PDF instructs the agent to reveal secrets	Treat retrieved text as untrusted data; test instruction-conflict cases.
Authorization bypass	Agent retrieves a restricted source by guessing its ID	Enforce policy inside adapters and retrieval, not in prompts alone.
Unsafe SQL	Generated query attempts write/delete or broad extraction	Allowlist schema/operations, validate statements, use read-only credentials and limits.
Secret exfiltration	Prompt asks model to print environment values	Never expose secrets to model context; test secret extraction attempts.
Tool misuse	Agent calls an unapproved URL or tool	Use explicit tool registry and external-source allowlist.
Malicious file	Document exploits parser or carries hostile content	Validate file type/size, isolate parsing where feasible and keep dependencies patched.
Citation fabrication	Model invents a page or source ID	Render citations only from validated evidence objects.
Cross-workspace leakage	Prior context influences another workspace query	Scope memory/context and retrieval to authenticated workspace.

23.1 Threat-handling rules

• The model cannot grant itself permission or override policy.

• Tool arguments are validated independently of model-generated text.

• Retrieved documents may contain instructions, but those instructions are never trusted as system policy.

• High-impact operations are read-only in the reference scope.

• Security tests are repeatable and recorded; a passing demo is not a substitute for a formal production assessment.

24. Privacy, Governance & Auditability

24.1 Audit event minimum

Record, at minimum, actor/user, timestamp, workspace/industry, query or query reference, agents invoked, sources accessed, retrieved evidence references, model/tool metadata where available, answer status, citations, grounding/conflict status, latency and errors. Apply data minimization and redact sensitive content according to policy.

24.2 Governance requirements

• Maintain source ownership and classification metadata.

• Document the source and version behind every material evidence item.

• Define how removed/replaced sources affect future retrieval and historical traces.

• Provide an auditable record of administrative changes to sources, roles and industry packs.

• Distinguish synthetic demonstration data from public data and any future customer data.

• Document retention/deletion decisions before the platform is used with real enterprise content.

24.3 Trust boundaries

The platform should not claim that a source is authoritative merely because it was retrieved. Source reliability depends on ownership, freshness, version, context and the use case. The UI should help users inspect those attributes rather than hide them behind a single trust score.

24.4 Human accountability

The system supports investigation and decision-making; it does not silently execute operational changes or replace the responsible business owner. Any future write-capable workflow requires separate approval, controls, audit and risk assessment.

25. Observability & Operational Diagnostics

25.1 Runtime signals

Signal	Minimum captured fields / purpose
End-to-end latency	Total duration and stage durations for representative queries.
Agent execution	Agent type, status, start/end time, duration and bounded error category.
Source retrieval	Source type/ID, result count, retrieval duration and failure status.
Tool calls	Tool name, validated arguments summary, outcome and timing; redact sensitive values.
Model usage	Provider/model identifier and token/usage metadata where available.
Quality signals	Evidence count, citation validation result, conflict status and insufficient-evidence status.
Reliability	Timeouts, retries, exceptions, source availability and application failure rate.

25.2 Operational requirements

• Use correlation IDs across API, orchestration, agents and source adapters.

• Keep user-facing error messages separate from internal diagnostic details.

• Bound retries and timeouts to avoid runaway execution.

• Make source/agent failures visible in the answer when they materially limit coverage.

• Avoid logging raw secrets or unnecessary sensitive payloads.

• Provide a repeatable local startup and demo smoke test.

25.3 Demo operations

The demo operator should have a known-good environment, preloaded synthetic data, a small set of validated scenarios and a recovery path if an external source is unavailable. If external retrieval is not essential to a particular scripted question, use an explicitly labeled synthetic external fixture rather than pretending a live lookup succeeded.

26. Evaluation Framework & Quality Metrics

Create an evaluation set of approximately 20–30 questions per industry pack. Include simple retrieval, multi-source questions, structured-data questions, cross-source reasoning, conflicting evidence, insufficient-evidence cases and multi-agent questions.

Dimension	What to measure	Evaluation method
Answer correctness	Conclusion aligns with available evidence	Human-reviewed expected answer or rubric
Grounding	Material claims supported by retrieved evidence	Claim-to-evidence review
Citation correctness	Citation actually supports attached claim	Citation verification against source
Retrieval quality	Relevant evidence found; irrelevant evidence minimized	Expected-evidence coverage and relevance review
Conflict handling	Disagreement surfaced rather than collapsed	Known conflict fixtures
Insufficient evidence	Answer qualifies/refuses when support is inadequate	Unanswerable questions with expected abstention
Latency	Observed response time	Record median and tail latency on defined environment
Failure rate	Tool/retrieval/orchestration/app failures	Run logs over repeated test set

26.1 Test set composition

- Single-source fact lookup.
- Multi-document synthesis.
- Structured metric calculation.
- Document + structured-data cross-check.
- External context with retrieval timestamp.
- Conflicting versions or mismatched time windows.
- Insufficient evidence / answerable only with missing source.
- Prompt injection and access-boundary adversarial cases.
- Follow-up query that relies on context without crossing workspace boundaries.

26.2 Reporting discipline

Report environment, dataset version, run date, sample size and scoring method. Measure new-demo latency independently. Do not reuse historical case-study metrics as if they describe this implementation.

27. Non-Functional Requirements

Area	Requirement for reference platform	Validation approach
Performance	Measure latency for representative scenarios; bound agent/source timeouts.	Record stage and end-to-end timings; report observed results.
Reliability	Recover from a non-critical source failure without fabricating success.	Inject timeout and adapter failure; inspect partial answer.
Security	Server-side authorization, least privilege and safe tool boundaries.	Automated tests plus focused manual security checks.
Maintainability	Small modules, typed contracts and clear service boundaries.	Code review and architecture walkthrough.
Configurability	Industry pack can be replaced without core rewrite.	Run same smoke scenarios under both packs.
Usability	Executive can grasp answer and evidence flow quickly.	Moderated 60–90 second comprehension test.
Accessibility	Keyboard-operable primary flow and non-color-only status.	Manual keyboard and contrast review.
Observability	Execution traces and errors correlated by request ID.	Inspect representative successful and failed runs.
Portability	Documented environment setup and configuration.	Clean setup from runbook on a fresh environment.

27.1 Performance target policy

The source PRD does not establish numeric service-level objectives for the new platform. Define initial targets only after baseline measurements on a documented environment. Any future target should specify workload, concurrency, data volume, model/provider, network conditions and percentile.

27.2 Scalability path

Keep adapters and orchestration replaceable so future deployments can introduce managed identity, durable queues, separate indexing workers, scalable vector storage, caching and production monitoring if validated by customer needs.

28. Testing Strategy & Release Gates

Test layer	Minimum expectation	Release evidence
Unit	Agent logic, retrieval utilities, validation, access checks and transformations	Automated suite and readable results
Integration	End-to-end source retrieval, orchestration, evidence flow and answer generation	Repeatable integration scenarios
Security	Authentication, authorization, injection, malicious documents, tool restrictions, secret exposure and unsafe data access	Recorded checks and remediations
Evaluation	Correctness, grounding, citations, conflicts, abstention and multi-source reasoning	Per-pack evaluation report
Regression	Previously passing critical scenarios remain passing	Versioned regression suite
Smoke	Clean startup and full demo path work	Runbook checklist
Configuration	Pack schema validation and industry switch	Both packs pass smoke checks

28.1 Release gates

1. Architecture and threat checklist reviewed before significant implementation.

2. Core query flow and source contracts pass integration tests.

3. Security controls are implemented server-side and tested.

4. Manufacturing evaluation set has been run and reviewed.

5. Retail & CPG pack passes the same shared-core smoke and quality checks.

6. Critical regressions and known failures are documented.

7. Demo script is repeatable and contains no unsupported performance claims.

28.2 Defect severity guidance

Block release for access-control bypass, secret exposure, fabricated citations presented as verified, cross-workspace leakage, unsafe structured-data execution, or failure to disclose material source limitations. Track cosmetic UI defects separately from trust and security defects.

29. Delivery Plan, Milestones & Dependencies

Phase	Outcome / checkpoint	Exit evidence
1. Architecture & planning	Architecture, boundaries, data flow, security assumptions, configuration map and CXO storyboard	Review sign-off on pre-coding submission
2. Core platform	Application shell, authentication, source management, configuration framework and basic query flow	Working query path and smoke test
3. Multi-agent intelligence	Router, document/structured/external agents, evidence validation and synthesis	Cross-source question works end-to-end
4. Enterprise controls	RBAC, source/workspace access, audit, observability and security cases	Control tests and trace inspection
5. Manufacturing pack	Synthetic/public data, documents, terminology, scenarios and evaluation set	Manufacturing demo and evaluation report
6. Retail & CPG validation	Second pack proving configuration reuse	Same core passes Retail smoke suite
7. Evaluation & hardening	Regression, security review, latency observations and cleanup	Quality and gap report
8. Sales packaging	Demo video, screenshots, architecture diagram, script and technical overview	Prospect-safe package reviewed

29.1 Mandatory pre-coding submission

• Architecture diagram and component boundaries.

• Configuration-versus-code map.

• Manufacturing data/source plan.

• CXO demo storyboard.

• Agent responsibilities and tool boundaries.

• Production-architecture versus demo-only gap assessment.

• Security threat checklist and evaluation approach.

29.2 Dependencies

Delivery depends on an agreed technology stack, approved model/provider access, sample synthetic/public source material, stakeholder review availability, and a chosen local or hosted demo environment. If a dependency is unavailable, document the fallback and label it accurately.

30. Deliverables & Definition of Done

30.1 Deliverables

• Working web application demonstrating Enterprise Knowledge Intelligence.

• Manufacturing / Industrial industry pack.

• Retail & CPG validation pack.

• Multi-agent orchestration and source-specific agents.

• Document, structured-data and controlled external retrieval.

• Evidence validation, traceability, conflict handling and insufficient-evidence behavior.

• Authentication, RBAC, access enforcement, audit and observability.

• Automated tests, evaluation datasets and results.

• Configuration-driven industry architecture and documentation.

• Architecture diagram and technical overview.

• 3–5 minute sales demo video and demo script.

• Prospect-safe screenshots/visual collateral.

• Setup, environment and runbook documentation.

30.2 Definition of done

Area	Done when
Functionality	A realistic multi-source business question is answered end-to-end.
Evidence	Important claims trace to sources; unsupported claims are qualified.
Multi-agent	Specialized agents are used appropriately and work is observable.
Security	Authentication, authorization and tool/source boundaries are server-side and tested.
Governance	Queries, agents, sources and key execution metadata are auditable.
Modularity	Manufacturing and Retail & CPG use the same core with configuration/data changes.
Quality	Unit, integration, security and evaluation tests exist; critical scenarios pass.
Performance	New-demo latency is measured and reported, not borrowed from another implementation.
UX	A non-technical executive can understand value and evidence flow quickly.
Sales readiness	The demo tells a clear story and shows how it maps to a prospect’s environment.

31. Risks, Assumptions & Open Questions

31.1 Risks and guardrails

Risk	Guardrail
Becoming a generic chatbot	Require multi-source questions, visible agents and evidence.
Overbuilding infrastructure	Build only what the demo needs while keeping clean production paths.
Hard-coded industry logic	Review configuration/code boundary before implementation.
AI-generated code defects	Human review, tests, security checks and ownership of committed code.
NDA/client leakage	Use only synthetic or approved public data and prospect-safe assets.
Hallucinated answers	Evidence validation, traceability and insufficient-evidence behavior.
False performance claims	Measure this implementation independently.
Weak architecture behind polished UI	Maintain clear service boundaries, authorization, logging and contracts.
Regulated-industry distraction	Start with Manufacturing; add sector-specific governance when pursuing those packs.

31.2 Working assumptions

- The first release is a demonstration/reference platform, not a certified production service.
- Synthetic/public data is sufficient to demonstrate cross-source reasoning.
- At least one structured source and one document source will be available for the first scenario.
- External intelligence may use a controlled source or explicitly labeled synthetic fixture.
- Model and infrastructure choices remain open if they satisfy the required boundaries.

31.3 Open decisions to resolve

- Which model/provider and embedding model are approved for the demo?
- Will structured data use CSV, PostgreSQL, or both in the first build?
- Which document formats and extraction pipeline are in scope?
- What identity/authentication mechanism will be used for the demo?
- What retention policy applies to query content and audit records?
- What deployment target and network restrictions apply?
- Who approves synthetic domain data and evaluates answer quality?

Decision discipline Unresolved items should be logged with an owner and decision date; do not silently convert assumptions into requirements.

32. AI-Assisted Development & Engineering Standards

AI-assisted coding is explicitly permitted where it accelerates delivery. It does not transfer engineering accountability from the implementer. Generated code must be understood, reviewed, tested and secured before it is committed.

32.1 Permitted uses

• Boilerplate and UI scaffolding.

• API scaffolding and repetitive adapters.

• Test-case generation and documentation drafts.

• Refactoring and debugging assistance.

• Implementation alternatives and code review prompts.

32.2 Required practices

• Understand every generated code path before committing it.

• Review dependency provenance, licensing and security implications.

• Test negative cases, timeouts and malformed inputs—not only the happy path.

• Document meaningful architectural decisions and trade-offs.

• Validate all generated SQL/tool access independently.

• Keep modules small, interfaces clear and configuration separate from core logic.

32.3 Never acceptable

• Copying code without understanding it.

• Hard-coding credentials or secrets.

• Disabling security controls to make a demo work.

• Trusting model-generated SQL without validation.

• Treating generated tests as proof of correctness.

• Using client data or screenshots without explicit authorization.

32.4 Engineering hygiene

Use version control, reproducible environment setup, explicit dependency versions where appropriate, safe configuration examples, useful error messages, code review and a documented test command. Remove unnecessary AI-generated complexity before release.

33. CXO Demo Storyboard & Sales Readiness

33.1 Recommended 3–5 minute flow

Time	Story / on-screen action
0:00–0:20	Show the business problem: critical knowledge is fragmented across documents, systems and external information.
0:20–0:40	Ask a realistic manufacturing CXO question requiring multiple source types.
0:40–1:30	Show the platform identifying relevant knowledge sources and specialized agents.
1:30–2:15	Show the consolidated answer with key findings and metrics.
2:15–2:45	Open evidence traceability and show where important claims came from.
2:45–3:15	Demonstrate a conflict or insufficient-evidence case.
3:15–3:45	Switch to Retail & CPG or inspect the configuration layer to prove reuse.
3:45–5:00	Explain how the architecture can map to a prospect's sources, terminology and business questions.

33.2 Demo operator checklist

• Confirm correct industry pack and source health.

• Use the validated question and known-good data snapshot.

• Verify citations open the expected source context.

• Show a pre-tested conflict or insufficient-evidence example.

• Demonstrate industry switching without editing core code.

• Avoid unmeasured claims about speed, accuracy, ROI or production readiness.

• Keep client identifiers and proprietary information out of screen captures.

33.3 Sales-safe claims

Say what the reference platform demonstrates: multi-source retrieval, specialized agents, evidence validation, source traceability and configuration reuse. Do not promise a specific operational ROI, accuracy level, latency, compliance certification or production capability unless it has been validated for the stated environment.

34. Appendices, Glossary & Final Product Statement

Appendix A — Glossary

Term	Definition
Enterprise Knowledge Intelligence	Business-facing capability that turns fragmented knowledge into contextual, traceable answers.
RAG	Retrieval-augmented generation: model output informed by retrieved source material.
Agent	A bounded software component that reasons over a task and uses approved tools.
Evidence item	A source-linked excerpt, metric or record used to support or qualify a claim.
Grounding	The degree to which an answer is supported by available evidence.
Industry pack	Configuration and data bundle for a particular industry/use case.
Prompt injection	Untrusted content attempting to override system or developer instructions.
Insufficient evidence	A state where the available sources do not support a reliable conclusion.

Appendix B — Representative question bank

• Manufacturing: What changed between maintenance procedure versions, and what evidence relates that change to downtime?

• Manufacturing: Which suppliers correlate with quality-impact events, and what corrective actions are documented?

• Retail: Why did Product X underperform despite increased promotion?

• Retail: Do sales and consumer research disagree, and how should the answer be qualified?

• Cross-domain: Which sources are stale, conflicting or insufficient to support the proposed conclusion?

Appendix C — Reference architecture statement

The logical architecture should preserve clear boundaries across experience, application, orchestration, intelligence, retrieval, governance, observability and configuration. Technology choices can vary; the responsibilities and security boundaries remain mandatory.

Final product statement

The promise	Enterprise knowledge does not have to remain fragmented across documents, databases and approved external sources. A governed, multi-agent knowledge intelligence layer can bring those sources together, retrieve relevant evidence, reason across it and return a traceable answer in a form business leaders can use. The implementation must demonstrate this credibly, safely and repeatedly across industries—not merely describe it.